From Double Integrals to Triple Integrals

A double integral adds up a function over a region in the plane: chop the region into small rectangles of area $\Delta A = \Delta x\,\Delta y$, sample f on each piece, add, and take a limit. A triple integral does the same thing one dimension up. The function is now $f(x, y, z)$, and the domain is a *solid* in $\mathbb{R}^3$.

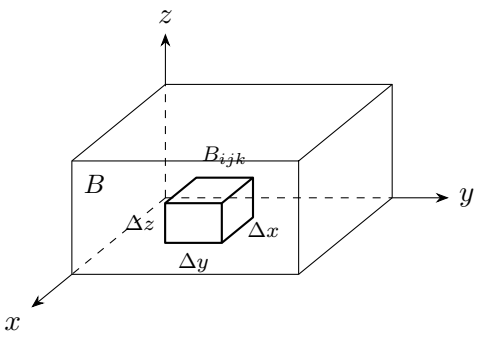
Start with a rectangular box $B = [a, b] \times [c, d] \times [e, f]$, the set of all points (x, y, z) with

$$a \le x \le b, \qquad c \le y \le d, \qquad e \le z \le f.$$

Divide $[a, b]$ into l pieces, $[c, d]$ into m pieces, and $[e, f]$ into n pieces. This cuts B into small sub-boxes B_{ijk}, each with volume

$$\Delta V = \Delta x\,\Delta y\,\Delta z.$$

Choose a sample point $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$ in each sub-box.



Definition (Triple Integral). The **triple integral** of $f(x, y, z)$ over the box B is

$$\iiint_B f(x, y, z)\,dV = \lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\,\Delta x\,\Delta y\,\Delta z$$

if this limit exists. If f is continuous on B, the limit exists and f is **integrable** on B.

What does a triple integral measure? The graph of $f(x, y, z)$ lives in four dimensions, so we can't picture the triple integral as "volume under a graph." It's better to think of f as a quantity spread through the solid, and of $f\,dV$ as the amount of it in one tiny piece.

- If $f(x, y, z) = 1$, each piece contributes its own volume, so $\iiint_E 1\,dV = V(E)$, the **volume** of the solid E.
- If $f(x, y, z)$ is a density (in g/cm^3), the triple integral is the total **mass** of the solid.

Triple Integrals over a Box

Just as with double integrals, we never compute a triple integral from the limit. We evaluate it as an **iterated integral**: integrate in one variable while holding the other two constant, then repeat from the inside out.

Fubini's Theorem for Triple Integrals. If f is continuous on the box $B = [a, b] \times [c, d] \times [e, f]$, then

$$\iiint_B f(x, y, z)\,dV = \int_e^f \int_c^d \int_a^b f(x, y, z)\,dx\,dy\,dz.$$

The iterated integral can be written in any of the **six** possible orders ($dx\,dy\,dz$, $dy\,dx\,dz$, $dz\,dy\,dx$, ...), and all give the same value.

Example 1. Evaluate $\iiint_B (x + yz^2)\, dV$, where $B = [0,1] \times [-1,2] \times [0,3]$. Use the order $dz\, dy\, dx$.

Shortcut. If the integrand is a product $g(x)\, h(y)\, k(z)$ and the region is a box, the triple integral splits into a product of three single integrals:

$$\iiint_B g(x)\, h(y)\, k(z)\, dV = \left(\int_a^b g(x)\, dx\right)\left(\int_c^d h(y)\, dy\right)\left(\int_e^f k(z)\, dz\right).$$

Practice 1. Let $B = [0,2] \times [0,1] \times [1,3]$.

(a) Evaluate $\iiint_B (x + yz)\, dV$ in any order you like.

(b) Use the shortcut to evaluate $\iiint_B 3x^2 yz\, dV$.

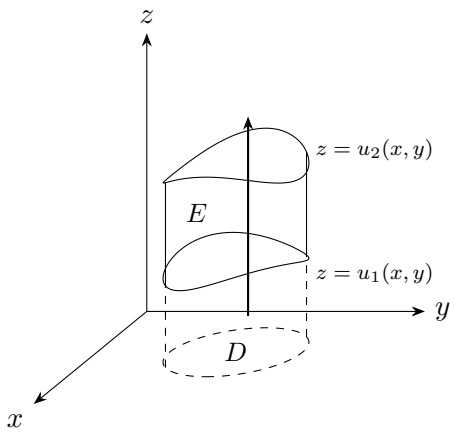
Triple Integrals over General Regions

Most solids aren't boxes. As with double integrals, we handle a general solid E by describing it with inequalities, and the inequalities become the bounds of integration.

Suppose E lies between a *bottom* surface $z = u_1(x, y)$ and a *top* surface $z = u_2(x, y)$. Let D be the **projection** of E onto the xy-plane, its "shadow" when light shines straight down. Then

$$E = \{(x, y, z) \mid (x, y) \in D,\ u_1(x, y) \le z \le u_2(x, y)\}.$$

To find the z-bounds, imagine a vertical arrow pointing up through E. It enters E through the bottom surface and leaves through the top surface.



Triple Integral over a General Region. If $E = \{(x, y, z) \mid (x, y) \in D,\ u_1(x, y) \le z \le u_2(x, y)\}$, then

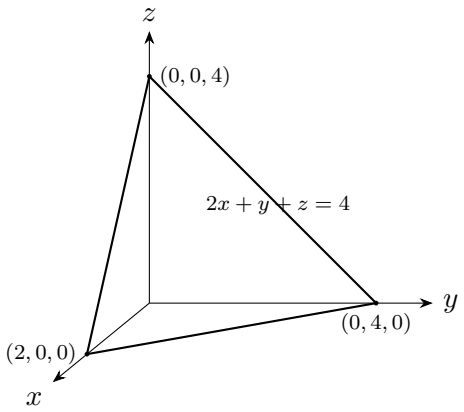
$$\iiint_E f(x, y, z)\, dV = \iint_D \left[\int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\, dz \right] dA.$$

The double integral over D is then set up exactly as before: as a Type I or a Type II region.

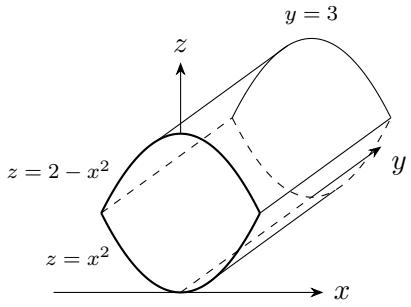
Setting up a triple integral.

1. **Inner bounds.** Find the bottom and top surfaces of E; they give the z-bounds.
2. **Shadow.** Find the projection D. Its boundary is often the curve where the top and bottom surfaces meet; set $u_1(x, y) = u_2(x, y)$ to find it.
3. **Outer bounds.** Describe D as a Type I or Type II region, just as for a double integral.
4. **Check.** The outer bounds are constants, the middle bounds involve only the outer variable, and the inner bounds involve at most the two outer variables.

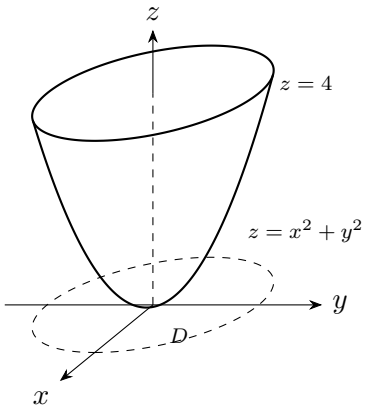
Example 2. Let E be the solid tetrahedron in the first octant bounded by the coordinate planes and the plane $2x + y + z = 4$. Evaluate $\iiint_E x\,dV$.



Example 3. Find the volume of the solid E that lies above the surface $z = x^2$, below the surface $z = 2 - x^2$, and between the planes $y = 0$ and $y = 3$.

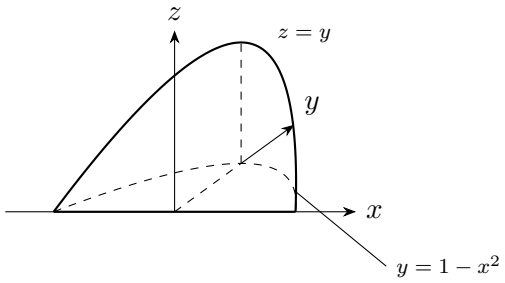


Example 4. Set up, but do not evaluate, $\iiint_E f(x,y,z)\,dV$, where E is the solid bounded below by the paraboloid $z = x^2 + y^2$ and above by the plane $z = 4$.



These bounds are correct, but the square roots make the integral painful to evaluate. We will finish this one with cylindrical coordinates, where the disk D is easy to describe.

Practice 2. Let E be the solid bounded below by the plane $z = 0$, above by the plane $z = y$, and on the sides by the parabolic cylinder $y = 1 - x^2$. Find the volume of E.



Projecting onto the Other Coordinate Planes

Nothing forces the shadow to fall on the xy-plane. Some solids are much easier to describe if we project them onto the yz-plane or the xz-plane and let the arrow run in the x- or y-direction.

Three ways to project.

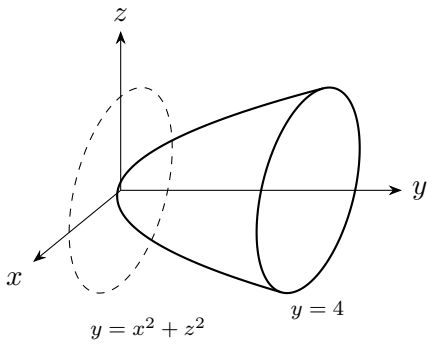
Shadow on the	Description of E	Triple integral
xy-plane	$(x,y) \in D, \quad u_1(x,y) \le z \le u_2(x,y)$	$\iint_D \int_{u_1(x,y)}^{u_2(x,y)} f \, dz \, dA$
yz-plane	$(y,z) \in D, \quad u_1(y,z) \le x \le u_2(y,z)$	$\iint_D \int_{u_1(y,z)}^{u_2(y,z)} f \, dx \, dA$
xz-plane	$(x,z) \in D, \quad u_1(x,z) \le y \le u_2(x,z)$	$\iint_D \int_{u_1(x,z)}^{u_2(x,z)} f \, dy \, dA$

A good choice: project onto the plane where the shadow is simplest, and let the arrow run in the direction where the solid has a single "entry" surface and a single "exit" surface.

Example 5. Let E be the solid bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$.

(a) Set up the volume of E by projecting onto the xy-plane. Why is this awkward?

(b) Set up the volume of E by projecting onto the xz-plane, then evaluate it.



Notice that the solid in Example 5 is the solid from Example 4 turned on its side, so the two solids have the same volume.

Changing the Order of Integration

Each projection gives two orders (one for each way to describe the shadow), for six orders in all. To switch orders, don't try to shuffle the bounds directly. Instead:

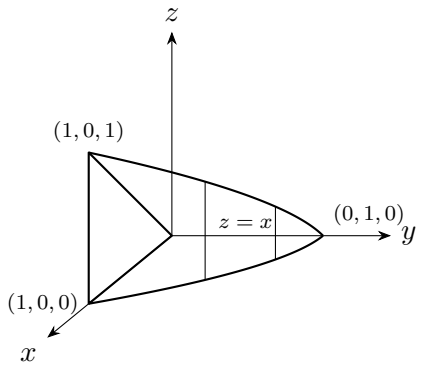
1. Read the inequalities that describe E off the bounds you were given.
2. Sketch E, or at least its shadows on the coordinate planes you need.
3. Re-describe E with the variables in the new order, from the outside in.

Example 6. Consider $\displaystyle\int_0^1 \int_0^{1-x^2} \int_0^x f(x,y,z)\,dz\,dy\,dx$.

(a) Describe the solid E with inequalities.

(b) Rewrite the integral in the order $dy\,dz\,dx$.

(c) Rewrite the integral in the order $dx\,dz\,dy$.



Practice 3. For the tetrahedron E of Example 2 (bounded by the coordinate planes and $2x+y+z=4$), write $\iiint_E f(x,y,z)\,dV$ as an iterated integral in each order.

(a) $dx\,dz\,dy$

(b) $dy\,dx\,dz$

Average Value of a Function of Three Variables

For a function of one variable, the average value is the integral divided by the length of the interval. For two variables, we divided by area. For three variables, we divide by volume.

Average Value. If f is integrable over a solid E with volume $V(E) > 0$, the **average value** of f over E is

$$f_{\text{ave}} = \frac{1}{V(E)} \iiint_E f(x,y,z)\,dV, \qquad \text{where } V(E) = \iiint_E 1\,dV.$$

Example 7. Find the average value of $f(x,y,z) = x$ over the tetrahedron E of Example 2.

Practice 4. Find the average value of $f(x,y,z) = y$ over the solid E of Practice 2. (You already found its volume.)

Answers

Practice 1. (a) 8 (b) 16, since $\left(\int_0^2 3x^2\,dx\right)\left(\int_0^1 y\,dy\right)\left(\int_1^3 z\,dz\right) = 8 \cdot \frac{1}{2} \cdot 4$.

Practice 2. $V = \displaystyle\int_{-1}^{1}\int_{0}^{1-x^2}\int_{0}^{y} 1\,dz\,dy\,dx = \frac{8}{15}$.

Practice 3. (a) $\displaystyle\int_0^4\int_0^{4-y}\int_0^{(4-y-z)/2} f\,dx\,dz\,dy$ (b) $\displaystyle\int_0^4\int_0^{(4-z)/2}\int_0^{4-2x-z} f\,dy\,dx\,dz$

Example 7 (if not finished in class). $\displaystyle\iiint_E x\,dV = \frac{8}{3}$ and $V(E) = \frac{16}{3}$, so $f_{\text{ave}} = \frac{1}{2}$.

Practice 4. $\displaystyle\iiint_E y\,dV = \frac{32}{105}$ and $V(E) = \frac{8}{15}$, so $y_{\text{ave}} = \frac{4}{7}$.